

AHP Decision Analysis Report

Car Selection

Goal: Select the optimal car considering safety, fuel efficiency, price, and comfort

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Executive Summary

Best Alternative: Honda CR-V (Score: 0.3510)

Number of Criteria: 4

Number of Alternatives: 4

Overall Consistency: Acceptable (CR = 0.0228)

Final Rankings

Rank	Alternative	Score	Percentage
1	Honda CR-V	0.3510	35.10%
2	Toyota RAV4	0.3302	33.02%
3	Mazda CX-5	0.1960	19.60%
4	Hyundai Tucson	0.1229	12.29%

Criteria Weights

Criterion	Weight	Percentage	Consistency Ratio
Safety	0.5654	56.54%	0.0076
Fuel Efficiency	0.2221	22.21%	0.0449
Comfort	0.1462	14.62%	0.0457
Price	0.0663	6.63%	0.0209

Alternative Scores by Criteria

Safety

Alternative	Score	Percentage
Toyota RAV4	0.3854	38.54%
Honda CR-V	0.3854	38.54%
Mazda CX-5	0.1422	14.22%
Hyundai Tucson	0.0871	8.71%

Fuel Efficiency

Alternative	Score	Percentage
Toyota RAV4	0.3107	31.07%
Honda CR-V	0.4393	43.93%
Mazda CX-5	0.1464	14.64%
Hyundai Tucson	0.1036	10.36%

Comfort

Alternative	Score	Percentage
Toyota RAV4	0.2533	25.33%
Honda CR-V	0.1791	17.91%
Mazda CX-5	0.4714	47.14%
Hyundai Tucson	0.0962	9.62%

Price

Alternative	Score	Percentage
Toyota RAV4	0.0939	9.39%
Honda CR-V	0.1404	14.04%
Mazda CX-5	0.2133	21.33%
Hyundai Tucson	0.5525	55.25%

Consistency Analysis

The Analytic Hierarchy Process includes consistency checking to ensure the reliability of judgments.

- **Consistency Ratio (CR) $\leq$ 0.1:** Acceptable consistency
- **Consistency Ratio (CR) $>$ 0.1:** Inconsistent judgments, review recommended

Overall Consistency Ratio: 0.0228

Methodology

This analysis used the Analytic Hierarchy Process (AHP), a multi-criteria decision analysis method developed by Thomas Saaty. The process involves:

1. **Hierarchy Construction:** Decomposing the decision into goal, criteria, and alternatives
2. **Pairwise Comparisons:** Comparing elements using Saaty's 1-9 scale
3. **Priority Derivation:** Calculating weights using the eigenvector method
4. **Consistency Verification:** Ensuring logical consistency of judgments
5. **Synthesis:** Combining weights to determine final priorities